

Admittance Control for Robotic-Assisted Tele-Echography

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Abstract—This paper discusses a new control approach for robotic-assisted tele-echography. An indirect force control (admittance control) establishes the contact dynamics between the echographic probe and the patient, outputting a compliant reference trajectory, used as reference for an inner motion controller. The motion control features computed torque control techniques in joint space, taking into account dynamic and kinematic parameters, as well as stochastic design through active observers (AOBs). Joint space control is driven by task space posture errors converted into joint velocities references. Experiments with a 7-DOF WAMTM manipulator have been carried out, where a human operator teleoperates an ultrasound probe in 6 Cartesian dimensions in typical probe/patient echographic interactions.

I. INTRODUCTION

Ultrasound examination is one of the most common and widespread medical imaging techniques. By pressing an ultrasound probe against the anatomical area under examination, the physician is provided with access in real time to the interior of the human body in a safe and noninvasive way. This intrinsically safe procedure together with a relatively low cost, specially when compared with other imaging procedures, makes ultrasound imaging an essential procedure for a broad group of medical specialties, providing access either to superficial systems such as muscles, tendons or vessels as well as to deeper systems like heart or liver. Being an important diagnosis tool, it is not a trivial procedure, requiring skilled and experienced physicians to correctly execute and diagnose the patient, which are not always available, specially in remote areas medical centers. A robotic-assisted tele-echography system presents a solution to this problem, allowing the physician to execute an echographic examination in a remotely placed patient, in a similar fashion to the standard procedure.

However, for a remote robotic-assisted tele-echographic system to be accepted and used by the medical community, the robot interacting with the patient has to accurately replicate physician's movements, while the device that the physician is handling must clearly reflect the forces that are exerted in the patient, enabling the physician with the same tactile feeling of an in-site procedure. Furthermore, the system should be versatile enough to be used by a wide range of medical specialties where echographic examination is a crucial

diagnosis tool, presenting a dynamic behavior in accordance with the examination nature, since contact forces between probe and patient are procedure specific. For instance, in an echocardiography (ultrasound examination of the heart), where the ultrasound probe has to be placed in a window between the ribs in the thoracic cavity, the required forces are greater in magnitude than, for instance, the echographic examination of the carotid artery. This clearly shows that robot control should be in accordance with the examination, assuming a stiff behavior in situations where high forces are needed, and a more compliant nature in procedures that require low interaction forces.

In this paper we will focus on robot dynamic control for echographic examination. Impedance control is one of the most common and reliable approaches to manage the contact dynamics, imposing a dynamic behavior of a mass-damper spring system between the end-effector and the environment, where a more or less compliant behavior can be established in accordance with the specific application [1]. However, the selection of proper impedance parameters that ensure a proper compliant behavior when in contact, might entail inadequate trajectory tracking in free space. A solution to this problem can be achieved by dealing with the motion control independently from impedance control [2]. An inner motion control loop can be used so that disturbance rejection is enhanced, tracking a trajectory computed by the impedance controller, instead of following the desired trajectory. The impedance controller becomes an admittance one, where desired position and force are inputs and a complaint position is the output.

This paper discusses a new control architecture for a robotic-assisted tele-echography system. Following the approach presented in [3], where a computed torque control in joint space driven by task space posture errors is presented, this approach extends [3] by introducing an indirect force control (admittance loop) to deal with position contact dynamics while improving the motion control by using the AOB framework to feature model-reference adaptive control.

The remaining of this paper is organized as follows. Section II presents the manipulator dynamic model and the system linearization. The interaction control strategy is explained in Section III while in Section IV the motion control architecture is discussed, including the AOB algorithm and the control design. Section V presents experimental results assessing the control performance and Section VI concludes the paper.

II. MANIPULATOR DYNAMIC MODEL

The manipulator dynamic model establishes the relation between robot motion and forces acting on it. Given a set of n

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generalized coordinates, q , describing the robot posture¹ and a set of generalized forces (torques), τ , doing work on the manipulator, the dynamic model can be formulated as

$$B(q)\ddot{q} + c(q, \dot{q}) + g(q) = \tau, \quad (1)$$

where $B(q) \in \mathbb{R}^{n \times n}$ is the manipulator inertia matrix (symmetric and positive definite), $c(q, \dot{q}) \in \mathbb{R}^n$ is the vector of Coriolis and centripetal torques and $g(q) \in \mathbb{R}^n$ is the term due to gravity torques.

A. Nonlinear Feedback Linearization

The manipulator defined by equation (1) represents a nonlinear system. With the identification of the dynamic model [5], it is possible to cancel the nonlinear terms through nonlinear feedback linearization. Let the torques acting on the manipulator be defined by

$$\tau = \tau_c - \tau_f - \tau_e, \quad (2)$$

where $\tau_c \in \mathbb{R}^n$ are actuator torques, $\tau_f \in \mathbb{R}^n$ the torques due to friction and $\tau_e \in \mathbb{R}^n$ are the torques arising from the interaction of the end-effector with the environment. Assuming that joint frictions could be neglected ($\tau_f \approx 0$) and defining the estimation of interaction torques $\hat{\tau}_e$ as

$$\hat{\tau}_e = J^T(q)\hat{f}_e, \quad (3)$$

where $\hat{f}_e \in \mathbb{R}^m$ is the environment force estimation acting on the end-effector and $J^T \in \mathbb{R}^{n \times m}$ is the Jacobian transpose matrix, the commanded torque τ_c can be defined as

$$\tau_c = \hat{c}(q, \dot{q}) + \hat{g}(q) + \hat{\tau}_e + \tau'_c, \quad (4)$$

where $\hat{c}(q, \dot{q}) \in \mathbb{R}^n$ and $\hat{g}(q) \in \mathbb{R}^n$ are proper estimations of $c(q, \dot{q})$ and $g(q)$. The system plant is reduced to

$$B(q)\ddot{q} = \tau'_c. \quad (5)$$

If τ'_c is set as

$$\tau'_c = \hat{B}(q)\alpha, \quad (6)$$

and $\hat{B}(q) \in \mathbb{R}^{n \times n}$ is a proper estimation of the inertia matrix, a new control variable $\alpha \in \mathbb{R}^n$ arises, representing a *resolved acceleration* in the joint space. With this approach, a linear and decoupled system is achieved, corresponding to a double integrator plant,

$$\ddot{q} = \alpha. \quad (7)$$

Equation (7) accounts for a perfect compensation of the terms in (1), being the overall system performance directly connected with the accuracy of dynamic parameter estimation. In this setup, the dynamic parameters were obtained through CAD models provided by the robot manufacturer. Joint frictions are seen as perturbations in the controller architecture while task space force estimation is obtained through force sensing.

¹The term posture denotes the pose (position and orientation) of all rigid bodies composing the robotic chain [4].

III. ADMITTANCE CONTROL

The compensation of contact forces in (3) makes the manipulator infinitely stiff, when seen from the environment standpoint, but for an application like robotic-assisted tele-echography, a compliant behavior is required. Impedance control is one solution to handle this problem, establishing a relation between contact forces and position errors, through an equivalent mechanical system whose dynamics are specified by a mass $M \in \mathbb{R}^{n \times n}$, a damping $D \in \mathbb{R}^{n \times n}$ and a stiffness matrix $K \in \mathbb{R}^{n \times n}$. Let $\nu_d \in \mathbb{R}^n$ be a generic desired position reference and $\nu_r \in \mathbb{R}^n$ a generic robot position, the impedance equation is given by

$$M(\ddot{\nu}_d - \ddot{\nu}_r) + D(\dot{\nu}_d - \dot{\nu}_r) + K(\nu_d - \nu_r) = f, \quad (8)$$

where $f \in \mathbb{R}^n$ is the contact force applied by the robot and measured by a force sensor, which allows to achieve a linear and decoupled impedance. In free space, the impedance controller is equivalent to an inverse dynamics controller. To provide good disturbance rejection with such controller, the stiffness gain K that corresponds to the proportional gain of a PD controller, should be kept high, which conflicts with a compliant behavior desired for tele-echography, where interaction forces should be limited (K should be low). This means that the selection of impedance parameters that ensure proper compliant behavior when contact arises, are not adequate to provide accurate trajectory tracking in free space.

A possible solution to this problem can be achieved by dealing with motion control independently from impedance control [2], [4]. The impedance controller is used to provide a suitable reference for the motion controller, in accordance with the desired interaction dynamics, while the motion controller is projected to present a high tracking performance of the compliant frame. With this approach, the impedance controller becomes an admittance controller where measured contact force together with the desired reference position ν_d are the inputs, while a compliant position ν_c becomes the output, used as the reference for the inner motion controller (admittance orientation control is not addressed in this paper).

A. Admittance Position Control

Let $p_d \in \mathcal{R}^3$ be the desired task space reference position and $p_c \in \mathcal{R}^3$ be the compliant task space position². The impedance equation is set to present a mass-damper-spring behavior for the position displacement when the end-effector exerts a force $f_f \in \mathcal{R}^3$ on the environment, as

$$M_p(\ddot{p}_d - \ddot{p}_c) + D_p(\dot{p}_d - \dot{p}_c) + K_p(p_d - p_c) = f_f, \quad (9)$$

where $M_p, D_p, K_p \in \mathcal{R}^{3 \times 3}$ are diagonal positive definite matrices. The reference for the motion control is given by computing p_c from (9) by means of numerical integration

² Vectors, matrices and quaternions are related to the frame specified by the superscript. The absence of superscript means that they are related to the base frame.

techniques. $\ddot{\mathbf{p}}_c$ and $\dot{\mathbf{p}}_c$ are computed by the Newmark Method [6] as

$$\begin{aligned} \ddot{\mathbf{p}}_{c,k} = & \frac{1}{\beta h^2} (\mathbf{p}_{c,k} - \mathbf{p}_{c,k-1}) \\ & - \frac{1}{\beta h} \dot{\mathbf{p}}_{c,k-1} - \left(1 - \frac{1}{2\beta}\right) \ddot{\mathbf{p}}_{c,k-1} \end{aligned} \quad (10)$$

and

$$\begin{aligned} \dot{\mathbf{p}}_{c,k} = & \frac{\gamma}{\beta h} (\mathbf{p}_{c,k} - \mathbf{p}_{c,k-1}) \\ & - \left(\frac{\gamma}{\beta} - 1\right) \dot{\mathbf{p}}_{c,k-1} - h \left(\frac{\gamma}{2\beta} - 1\right) \ddot{\mathbf{p}}_{c,k-1} \end{aligned}, \quad (11)$$

where h is the sampling time and k is the discrete time instant. Substituting (10) and (11) in (9), while setting $\beta = \frac{1}{4}$ and $\gamma = \frac{1}{2}$ (trapezoidal rule [6]), $\mathbf{p}_c$ is then given by

$$\begin{aligned} \mathbf{p}_{c,k} = & \left(\frac{4}{h^2} \mathbf{M}_p + \frac{2}{h} \mathbf{D}_p + \mathbf{K}_p \right)^{-1} \\ & \cdot \left(\mathbf{M}_p \left(\frac{4}{h^2} \mathbf{p}_{c,k-1} + \frac{4}{h} \dot{\mathbf{p}}_{c,k-1} - \ddot{\mathbf{p}}_{c,k-1} + \ddot{\mathbf{p}}_{d,k} \right) \right. \\ & + \mathbf{D}_p \left(\frac{2}{h} \mathbf{p}_{c,k-1} + \dot{\mathbf{p}}_{c,k-1} + \dot{\mathbf{p}}_{d,k} \right) \\ & \left. + \mathbf{K}_p \mathbf{p}_{d,k} - \mathbf{f}_f \right) \end{aligned} \quad (12)$$

IV. MOTION CONTROL ARCHITECTURE

The inner motion control loop follows the approach of the architecture presented in [3], where a joint space velocity control driven by task space posture errors is presented, merging both joint and task space formalisms into a single architecture. The two loop approach allows to track a pose reference in the outer motion control loop, while the inner velocity loop enables to control each joint individually in a decoupled fashion. Fig. 1 shows the control architecture.

A. Task Space Posture Control

Let $\dot{\mathbf{x}}_p \in \mathbb{R}^6$ be task space velocity reference of the motion controller

$$\dot{\mathbf{x}}_p = \begin{bmatrix} \dot{\mathbf{p}}_p \\ \dot{\mathbf{w}}_p \end{bmatrix}, \quad (13)$$

composed by linear and angular task space velocities references $\dot{\mathbf{p}}_p \in \mathbb{R}^3$ and $\dot{\mathbf{w}}_p \in \mathbb{R}^3$, respectively. In [3] we proposed to compute $\dot{\mathbf{x}}_p$ from the task space error $\mathbf{x}_e \in \mathbb{R}^6$ as

$$\dot{\mathbf{x}}_p = \mathbf{I} \mathbf{x}_e, \quad (14)$$

with

$$\mathbf{x}_e = \begin{bmatrix} \Delta \mathbf{p}_{cr} \\ \boldsymbol{\epsilon}_{dr} \end{bmatrix}, \quad (15)$$

where $\mathbf{I} \in \mathbb{R}^{6 \times 6}$ is the identity matrix and $\Delta \mathbf{p}_{cr} \in \mathbb{R}^3$ and $\boldsymbol{\epsilon}_{dr} \in \mathbb{R}^3$ are task space position and orientation error, respectively. The position error is given by

$$\Delta \mathbf{p}_{cr} = \mathbf{p}_c - \mathbf{p}_r, \quad (16)$$

where $\mathbf{p}_c \in \mathbb{R}^3$ and $\mathbf{p}_r \in \mathbb{R}^3$ are compliant and robot end-effector task space positions, respectively. The mutual

orientation between desired $\mathcal{Q}_d$ and robot orientation $\mathcal{Q}_r$ (using an unit quaternion representation [7]), expressed in the robot frame ${}^r\mathcal{Q}_{dr} = \{\eta_{dr}, {}^d\boldsymbol{\epsilon}_{dr}\}$ is given as

$${}^r\mathcal{Q}_{dr} = \mathcal{Q}_r^{-1} * \mathcal{Q}_d, \quad (17)$$

being described in the base frame by

$$\begin{aligned} \mathcal{Q}_{dr} &= \mathcal{Q}_r * {}^r\mathcal{Q}_{dr} * \mathcal{Q}_r^{-1} \\ &= \mathcal{Q}_d * \mathcal{Q}_r^{-1}, \end{aligned} \quad (18)$$

where the symbol $*$ denotes quaternion multiplication. Knowing that $\mathcal{Q}_{dr} = \{1, \mathbf{0}\}$ only if the desired frame is aligned with the robot frame ($\mathcal{Q}_d = \mathcal{Q}_r$), the orientation error can be defined in a minimal orientation representation (three parameter representation) as [4]

$$\boldsymbol{\epsilon}_{dr} = \eta_r \boldsymbol{\epsilon}_d - \eta_d \boldsymbol{\epsilon}_r - \mathbf{S}(\boldsymbol{\epsilon}_d) \boldsymbol{\epsilon}_r, \quad (19)$$

The task space velocity reference $\dot{\mathbf{x}}_p$ is mapped into the joint space through the differential kinematics equation

$$\dot{\mathbf{q}}_p = \mathbf{J}^\dagger(\mathbf{q}) \dot{\mathbf{x}}_p, \quad (20)$$

where $\mathbf{J}^\dagger(\mathbf{q}) \in \mathbb{R}^{m \times n}$ is the Moore-Penrose pseudo-inverse of the Jacobian matrix. $\dot{\mathbf{q}}_p$ is the joint space counterpart of $\dot{\mathbf{x}}_p$, used to compute the reference for the inner joint velocity control.

B. Joint Space Velocity Control

The resolved acceleration $\boldsymbol{\alpha}$ is computed from a model reference adaptive control over the joint space velocity error. The joint velocity reference $\dot{\mathbf{q}}_c$ is computed by scaling $\dot{\mathbf{q}}_p$ by a proportional control gain $\mathbf{K}_1$,

$$\dot{\mathbf{q}}_c = \mathbf{K}_1 \dot{\mathbf{q}}_p. \quad (21)$$

From this point, the control analysis is done for each manipulator joint individually, being controlled by an AOB.

1) *System Plant*: The system plant of the velocity controlled manipulator (Fig. 2) is given by

$$G(s) = \frac{e^{-sT_d}}{s}, \quad (22)$$

where T_d is the system delay, arising mainly due to signal processing. The equivalent temporal representation is given by

$$\dot{y}(t) = \alpha(t - T_d) \quad (23)$$

where α is the plant input (joint acceleration) and $y(t)$ is the plant output (joint velocity). Defining $\mathbf{x}_{r1}(t) = y(t)$, (23) can be written in a state-space representation as

$$\begin{cases} \dot{\mathbf{x}}_r(t) = \mathbf{A} \mathbf{x}_r(t) + \mathbf{B} u(t - T_d) \\ \mathbf{y}(t) = \mathbf{C} \mathbf{x}_r(t) \end{cases}, \quad (24)$$

with

$$A = 0 \quad B = 1 \quad \text{and} \quad C = 1.$$

Discretizing (24) with sampling time h [8], the equivalent discrete time system becomes

$$\begin{cases} \mathbf{x}_{r,k} = \boldsymbol{\Phi} \mathbf{x}_{r,k-1} + \boldsymbol{\Gamma} u_{k-1} \\ \mathbf{y}_k = \mathbf{C}_r \mathbf{x}_{r,k} \end{cases}, \quad (25)$$

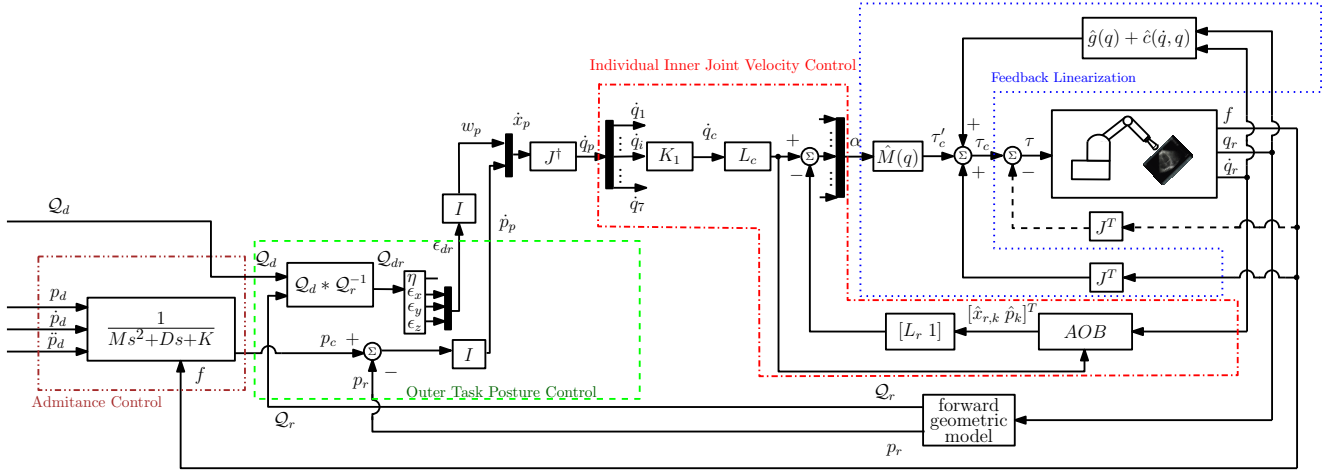


Fig. 1: Proposed control architecture for robot-assisted tele-echography. An admittance control loop provides compliant motion references for the motion control loop. The motion control architecture has a velocity model-reference adaptive control (AOB) in the joint space, driven by task space posture errors. Velocities from task space are mapped into the joint space following an inverse kinematics approach.

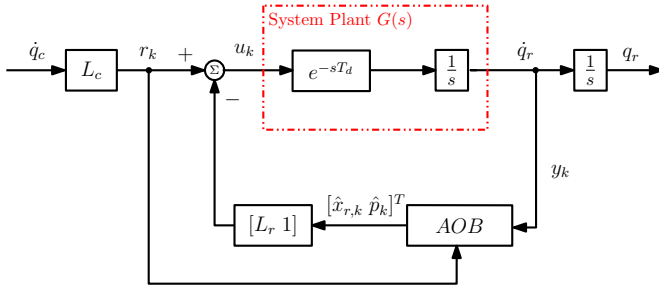


Fig. 2: Joint velocity control with an AOB in the loop, controlling each joint individually.

with

$$T_d = (d-1)h + \tau', \quad 0 < \tau' \leq h, \quad (26)$$

$$\mathbf{x}_{r,k} = [x_k \ u_{k-d} \ \dots \ u_{k-2} \ u_{k-1}]^T \quad (27)$$

$$\Phi_r = \begin{bmatrix} \Phi_1 & \Gamma_1 & \Gamma_0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 1 \\ 0 & 0 & 0 & \dots & 0 \end{bmatrix}, \quad (28)$$

$$\Gamma_r = [0 \ 0 \ \dots \ 0 \ 1]^T, \quad (29)$$

$$C_r = [1 \ 0 \ \dots \ 0 \ 0]. \quad (30)$$

Φ_1 , Γ_1 and Γ_0 are given by

$$\Phi_1 = e^{A h}, \quad (31)$$

$$\Gamma_1 = e^{A(h-\tau')} \int_0^{\tau'} e^{A \lambda} d\lambda B \quad (32)$$

$$\Gamma_0 = \int_0^{h-\tau'} e^{A \lambda} d\lambda B \quad (33)$$

2) *AOB Design:* An AOB (Active Observer) is a state space observer that reformulates the Kalman filter framework in order to accomplish model-reference adaptive control [9]. Resorting to an explicit pole placement approach, the AOB uses the stochastic design of the Kalman filter to reflect the uncertainties of system model and output measurements. An extra state (active state) p_k , is employed to estimate an equivalent disturbance referred to the system input due to unmodeled terms, such as higher order dynamics, parameter mismatches, unknown disturbances and noise. Controlling the system of (25) through an observer state feedback and inserting p_k and $\hat{p}_k$ in the loop, the system can be represented by

$$\begin{bmatrix} \mathbf{x}_{r,k} \\ \mathbf{p}_k \end{bmatrix} = \begin{bmatrix} \Phi_r & \Gamma_r \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \mathbf{x}_{r,k-1} \\ \mathbf{p}_{k-1} \end{bmatrix} + \begin{bmatrix} \Gamma_r \\ 0 \end{bmatrix} \mathbf{u}'_{k-1} + \boldsymbol{\xi}_k \quad (34)$$

$$\mathbf{y}_k = C_a \begin{bmatrix} \mathbf{x}_{r,k-1} \\ \mathbf{p}_{k-1} \end{bmatrix} + \boldsymbol{\eta}_k \quad (35)$$

with

$$\mathbf{u}'_{k-1} = \mathbf{r}_{k-1} - [L_r \ 1] \begin{bmatrix} \mathbf{x}_{r,k-1} \\ \hat{\mathbf{p}}_{k-1} \end{bmatrix}. \quad (36)$$

$$C_a = [C_r \ 0]$$

$\mathbf{r}_{k-1}$ is the input reference and L_r is the state feedback gain matrix used to set the closed loop system response. $\boldsymbol{\xi}_k$ and $\boldsymbol{\eta}_k$ represent the stochastic inputs due to, respectively, model and measure uncertainties. The state estimate of (34) is based on the desired closed loop ($p_k = \hat{p}_k$ and $\mathbf{x}_{r,k} = \hat{\mathbf{x}}_{r,k}$)

$$\begin{bmatrix} \hat{\mathbf{x}}_{r,k} \\ \hat{\mathbf{p}}_k \end{bmatrix} = \begin{bmatrix} \Phi_r - \Gamma_r L_r & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \hat{\mathbf{x}}_{r,k-1} \\ \hat{\mathbf{p}}_{k-1} \end{bmatrix} + \begin{bmatrix} \Gamma_r \\ 0 \end{bmatrix} \mathbf{r}_{k-1} + K_k (\mathbf{y}_k - \hat{\mathbf{y}}_k) \quad (37)$$

with

$$\hat{\mathbf{y}}_k = C_a \left(\begin{bmatrix} \Phi_r - \Gamma_r L_r & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \hat{\mathbf{x}}_{r,k-1} \\ \hat{\mathbf{p}}_{k-1} \end{bmatrix} + \begin{bmatrix} \Gamma_r \\ 0 \end{bmatrix} \mathbf{r}_{k-1} \right). \quad (38)$$

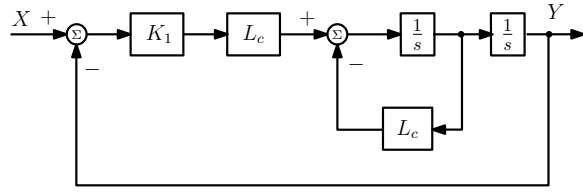


Fig. 3: Simplified motion control architecture.

The Kalman gain K_k reflects the uncertainty of each state, being a function of ξ_k and η_k [9]

$$K_k = P_{1k} C_a^T [C_a P_{1k} C_a^T + R_k]^{-1} \quad (39)$$

with

$$P_{1k} = \Phi_n P_{k-1} \Phi_n^T + Q_k \quad (40)$$

$$P_k = P_{1k} - K_k C_a P_{1k} \quad (41)$$

where Φ_n is the augmented state transition matrix

$$\Phi_n = \begin{bmatrix} \Phi_r & \Gamma_r \\ 0 & 1 \end{bmatrix}, \quad (42)$$

$R_k = E\{\eta_k \eta_k^T\}$ and $Q_k = E\{\xi_k \xi_k^T\}$ are, respectively, the measurement and the system noise matrix. Q_k has the form

$$Q_k = \begin{bmatrix} Q_{x_{r,k}} & 0 \\ 0 & Q_{p_k} \end{bmatrix} \quad (43)$$

P_k is the mean-square error matrix and its initial value reflect the uncertainty associated with the state estimation. Its initial value should not be lower than the initial value of Q_k . Fig. 2 shows the velocity control loop with an AOB.

C. Motion Control Design

The motion control is designed to provide a high tracking performance of the compliant frame, enforcing disturbance rejection. L_r is specified by the well known Ackerman's formula in accordance to the velocity loop time constant τ_v , being the poles due to the delay T_d mapped in $z = 0$ (discrete time control). A fast dynamic response is pursued by setting a small τ_v .

1) *Transient analysis*: Let's neglect for now the influence of $J^\dagger(q)$ and T_d . The pole placement gain reduces to L_c (first order system for the velocity loop), which is the first element of L_r . For a perfect perturbation cancellation ($p_k = \hat{p}_k$), the closed loop transfer function of the motion controller can be approximated by a second order system, with transfer function given by

$$G(s) = \frac{Y(s)}{X(s)} \approx \frac{K_1 L_c}{s^2 + L_c s + K_1 L_c}, \quad (44)$$

with position constant time given by

$$\tau_p = \frac{1}{\zeta \omega_n}. \quad (45)$$

Fig. 3 shows the simplified control architecture. Since a standard second-order system is given by

$$G(s) = \frac{\omega_n}{s^2 + 2\zeta\omega_n s + \omega_n^2}, \quad (46)$$

where ω_n is the natural underdamped frequency and ζ is the damping factor, from (44) and (46), and using (45), K_1 is equal to

$$K_1 = \frac{L_c}{2^2 \zeta^2}. \quad (47)$$

With the presence of $J^\dagger(q)$ in the control law, the decoupled joint control is no longer achievable (since the $J^\dagger(q)$ is a non diagonal matrix and varies with robot posture). However, the decoupled architecture provides a good "initial guess" for motion control gain design. After this initial design, the gains are adjusted to provide an approximately critically damped behavior ($\zeta = 1$) in the robot position, since overshoots/undershoots in motion control are typically undesired.

2) *Stochastic Design Strategy*: The observer gain K_k is only defined by the relation between the stochastic matrices Q_k and R_k , and not by their absolute value. Following a normalization strategy for R_k ($R_k = 1$), the estimation strategy and the K_k dynamics is set by designing Q_k in order to reflect the uncertainty of the states [10]. Setting Q_k with a relatively small value with respect to R_k , means that a model based strategy is followed, while a relatively high value of Q_k accounts for a sensor based approach. For the tele-echography application, a sensor based strategy proved to be adequate, allowing to accurately track the compliant reference.

V. EXPERIMENTS

The developed architecture is assessed by performing typical echographic motions, both in free space and contact, in a teleoperation scenario. Motion commands generated by a Phantom Desktop haptic device control the medical robot, a torque controlled 7-DOF WAM™ lightweight anthropomorphic robot, with a JR3 force sensor and an ultrasound probe attached at the end-effector. The compliant behavior generated by the admittance loop is assessed for 3D Cartesian position dimensions, while orientation control is analyzed in free space.

A. Control Design Setup

Although the WAM™ globally presents low frictions, the last three joints (specially the last one), due to their low inertia, present non negligible friction effects. Since friction is not taken into account in the dynamic model, its detrimental effect is tackled in the control design, by both the active state p_k , where frictions are seen as a perturbation, and by setting higher control gains for the last three joints. Table I summarizes control gains. The control sampling time is $h = 1.2[\text{ms}]$ and the system delay is $T_d \approx h$ (obtained experimentally). For the AOB stochastic design, all joints present the same sensor based approach,

$$Q_k = \begin{bmatrix} 10^{-2} & 0 & 0 \\ 0 & 10^{-12} & 0 \\ 0 & 0 & 10^{-6} \end{bmatrix},$$

which entails the following steady state Kalman gain

$$K_k = [9.51 \times 10^{-2} \quad 6.34 \times 10^{-4} \quad 6.34 \times 10^{-4}]^T.$$

TABLE I: Motion Control Design Setup

	τ_v	K_1	L_r
q_1	0.014	17.11	$[68.45, 8.21 \times 10^{-2}]$
q_2	0.014	17.11	$[68.45, 8.21 \times 10^{-2}]$
q_3	0.014	17.11	$[68.45, 8.21 \times 10^{-2}]$
q_4	0.014	17.11	$[68.45, 8.21 \times 10^{-2}]$
q_5	0.010	23.56	$[94.23, 1.13 \times 10^{-1}]$
q_6	0.010	23.56	$[94.23, 1.13 \times 10^{-1}]$
q_7	0.002	94.00	$[376.00, 4.51 \times 10^{-1}]$

B. Experimental Results

Two different admittance designs are tested. The first one presents a stiffer design, suitable for applications requiring high interaction forces, while the second one is more compliant, which is adequate for applications requiring limited forces. The admittance gains for the stiff design are:

$$K_{p1} = \begin{bmatrix} 1000 & 0 & 0 \\ 0 & 1000 & 0 \\ 0 & 0 & 1000 \end{bmatrix}, D_{p1} = \begin{bmatrix} 50 & 0 & 0 \\ 0 & 50 & 0 \\ 0 & 0 & 50 \end{bmatrix},$$

while for the compliant design are

$$K_{p2} = \begin{bmatrix} 500 & 0 & 0 \\ 0 & 500 & 0 \\ 0 & 0 & 500 \end{bmatrix}, D_{p2} = \begin{bmatrix} 100 & 0 & 0 \\ 0 & 100 & 0 \\ 0 & 0 & 100 \end{bmatrix},$$

The mass gains are equal in both designs, based on the diagonal elements of the robot task space mass matrix

$$A = (J(q)B(q)^{-1}J(q)^T)^{-1}, \quad (48)$$

given as

$$M_{p1} = M_{p2} = \begin{bmatrix} 0.1 A_{1,1} & 0 & 0 \\ 0 & 0.1 A_{2,2} & 0 \\ 0 & 0 & 0.1 A_{3,3} \end{bmatrix}.$$

1) *Experiments Description:* Since the nature of experiments involve contact with a human, repeatability is difficult. However, experiments are similar as possible and can be summarized as follows. Each experiment starts with the echographic probe in free space, until it reaches contact with the patient so that contact forces mainly arise in the z-axis. The first contact happens in the abdominal area (complaint surface) and the second one in the thoracic cavity (stiff surface), being the probe moved in free space between these two contact points. No force feedback is provided to the human operator.

2) *Discussion:* Figs. 4 and 5 present experimental results. In the first experiment with a stiff design, the contact in abdominal area arises around 18[s], and the contact with the thoracic cavity is around 35[s]. Results show that for a stiffer admittance design, a stable behavior is achieved in contact with both surfaces, as can be seen in the reduced force and position oscillations (see Figs. 4a and 5c). In the second experiment with a compliant design, the first contact arises around 10[s] and the second around 22[s]. Results show that similar position mismatches in both admittance designs result in reduced interaction forces in the compliant one,

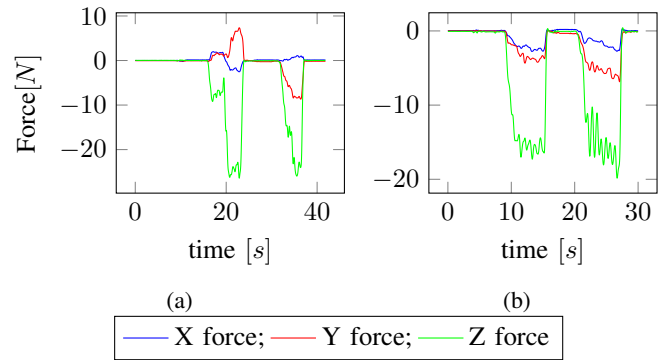


Fig. 4: Force exerted by the robot on the patient. (a) Stiff design. (b) Compliant design.

which emphasizes the indirect force control behavior of the admittance loop (see Figs. 5c, 5f, 4a and 4b). However, while the first contact (compliant surface) in the compliant design result in reduced force oscillations, the contact with stiffer surfaces (thoracic cavity) results in larger force oscillations (see Fig. 4b). Further experiments (not discussed in this paper) show that, when there is a big mismatch between admittance and environment stiffness, the system becomes unstable. Increasing the admittance damping leads to a stable behavior, however a large peak force occurs when contact takes place and large position errors arise when changing from contact to free space motion (the damping gain acts like the integrative gain in a PI controller).

3) *Free Space Behavior:* Position tracking in free space can be assessed in Fig. 5 by analyzing motion behavior before contact. Fig. 6 presents orientation tracking. Results show good pose tracking. For the orientation, small orientation error can be noticed, which can be explained due to coupling effects not properly canceled in feedback linearization and also due to the presence of $J^\dagger(q)$ in the control law.

VI. CONCLUSION

This paper has discussed the concept of indirect force control in the robotic-assisted tele-echography framework. An admittance control loop establishes the dynamic interaction between ultrasound probe and patient, providing a suitable compliant reference for motion control. The motion controller, grounded on joint space computed torque control techniques and boosted by stochastic design through Kalman active observers, enables good tracking performances of compliant references. Two admittance designs (stiff and a compliant), have been tested in typical robotic-assisted tele-echography scenario. Experimental results have shown that similar position mismatches in both designs lead to reduced interaction forces in the compliant design, validating the approach. Experiments also have shown that for significant mismatches between admittance and environment stiffness, higher damping gains are required for stable interaction, leading to undesirable behaviors for a robotic-assisted tele-echography system.

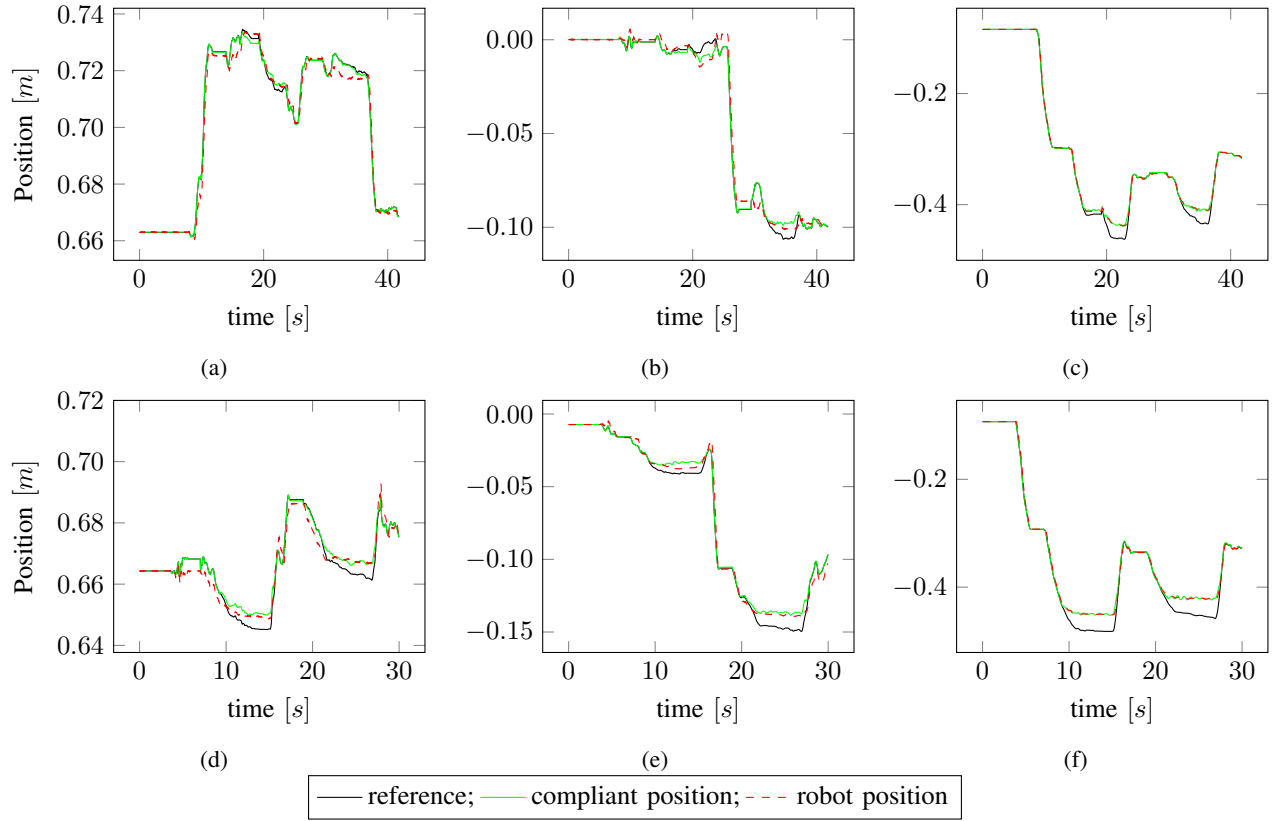


Fig. 5: Cartesian position tracking for the two admittance designs. (a), (b) and (c) are respectively X , Y and Z tracking for the stiff design and (d), (e) and (f) are compliant design counterparts.

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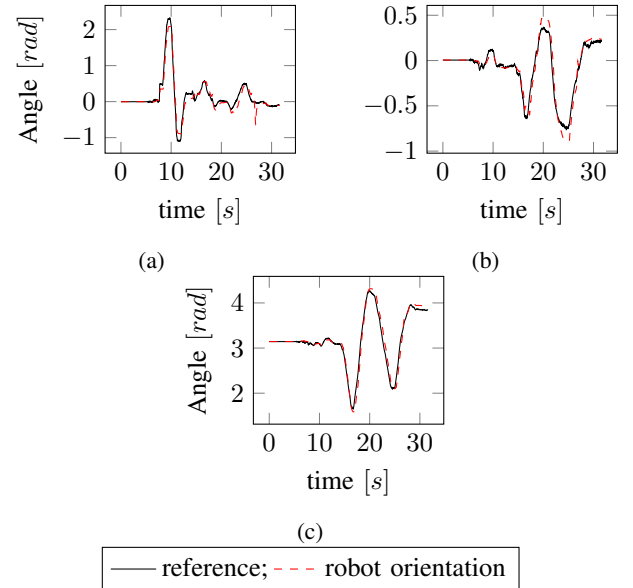


Fig. 6: Orientation tracking in free space. The orientation is represented by Roll, Pitch and Yaw angles. Roll angle $\in [-\frac{\pi}{2}, \frac{3\pi}{2}]$, Yaw $\in [0, 2\pi]$ and Pitch $\in [-\pi, \pi]$. (a) Roll tracking. (b) Pitch tracking. (c) Yaw tracking.